

A brief history of the meter, the SI-unit for length measurement

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This text is a synthesis of several Wikipedia articles on metrology.

Naturally, people who had to measure lengths have always used some unit of measurement or standard. Such a standard can, in principle, be chosen freely, for use within a limited scope. And initially, that was indeed common practice. The numerous old units of length (yard, ell, palm, foot, cubit, ...) ¹ illustrate this. And as long as measurements (i.e., the numerical values) can be traced back to the same standard, there is no problem.

The Scientific Revolution of the 16th and 17th centuries in Europe, was an irreversible break between ancient thought and modern science. The rapid accumulation of knowledge led, among other things, to the realization that the proliferation of measurement units, in particular for length measurements, should be countered.

The scientific developments of the era led to the recognition that it is essential to have precise and reproducible experiments. Also the internationalization of the natural sciences contributed to the realization that an internationally agreed system of measurement units would be at the benefit of the scientific exchanges.

It became gradually clear that measurement units should be truly constant in time and space. For the precision demanded by scientific experiments, the existing, artifact based standards, were insufficient. Even if such standards would be preserved and used with great care, they would become subject to damage or wear, affecting their value. Hence attempts were made to link measurement units to universal properties that were believed to be truly constant, yielding the same outcome for any observation thereof.

This eventually led to the proposal by the French Academy of Sciences in 1790 to define the unit of length, the meter, as being equal to one ten-millionth of the distance from the North Pole to the Equator (a quadrant of the Earth's circumference). This was to be measured along the meridian passing through the center of Paris Observatory. At that time, it was believed that this distance is fundamentally constant, leading to a consistent outcome of whoever wants to undertake the effort of observing it. As an aside, an earlier definition based on the second pendulum ² was rejected precisely due to the fact that the resulting meter would change with gravitational variations.

To determine the actual value of the meter, an expedition was set up in 1791 under the supervision of Jean-Baptiste Delambre and Pierre Méchain to precisely measure a part of the Paris meridian, namely that between the belfry of Dunkirk and the Montjuic castle of Barcelona. Based on their findings ³ the distance of the North Pole to the equator was extrapolated, taking into account the fact

¹ As can be seen, these ancient standards use the human body as starting point. Not a very precise foundation given the substantial variation in the dimensions of bodies.

² The relation between the period T and the length l of a gravitational pendulum is given by $T = 2\pi \sqrt{\frac{l}{g}}$ or equivalently $l = g \left(\frac{T}{2\pi}\right)^2$ with g the gravity acceleration, being about 9,81 m/s². A second pendulum has a period of exactly 2 seconds, 1 second per swing in any direction. Consequently such a pendulum has a theoretical length of 994 mm. This would enable the practical introduction of a consistent unit of length, were it not that g is not constant for every location on Earth.

³ Therefore the earth's polar circumference is (almost) equal to 40 000 km. So, this round figure is not a coincidence, but a direct consequence of the definition of the meter.

- by that time already well established - that the earth is not a true sphere, but somewhat flattened at the poles.

Evidently, this new definition of the meter could not simply be applied in practice immediately.⁴ Therefore, when the result of the expedition was finally established, the 'Mètre des Archives' – a platinum bar with a length precisely corresponding to the new definition, was created and designated by law as the primary standard meter (1799).⁵ Precise copies of this artifact would then be used to spread the standard geographically. One such copy was brought to the US in 1805.

Although the meter was adopted as unit of length for some important geographic projects, such as the US Coast Survey and the topographic mapping of Switzerland, it lasted till 1875 to adopt the meter as the international scientific unit of length. This was laid down in a treaty, the so-called Meter Convention, signed by 17 countries.⁶ The treaty also created the International Bureau of Weights and Measures (BIPM⁷), an intergovernmental organization, under the authority of the General Conference on Weights and Measures (CGPM⁸). The BIPM (meanwhile grown to 64 member states) still has the mandate to provide the basis for a single, coherent system of measurements throughout the world, traceable to the International System of Units (SI). The CGPM meets every 4 years and is the supreme authority validating the proposals in respect to changes to the SI.

The first meeting of the CGPM took place in September 1889. It was agreed to replace the original Mètre des Archives (and its copies) by an improved version, called international prototype meter. The improvements consisted of a new alloy, making it harder than pure platinum, a special X-shaped cross section (so called "Tresca section") to minimize the effects of torsional strain during length comparisons and the addition of two fine lines precisely marked at a distance of one meter, to make any wear at the ends of the bar irrelevant.⁹ The construction of the prototype meter was at the limit of the technology of the time. With some difficulty a lot of 30 bars could be produced to the required specifications. One of these, No. 6, was determined to be identical in length to the Mètre des Archives, and was designated the international prototype meter. The other bars, duly calibrated against the international prototype, were distributed to the signatory nations of the Meter Convention for use as national standards. For example, Belgium received No. 2 and the United States received No. 27. As bar lengths vary with temperature, precise measurements required known and stable temperatures and could even be affected by a scientist's body heat, so standard meters were provided with precise thermometers.

During the 6th meeting of the CGPM in 1921, the scope of the Meter Convention was extended to accommodate all physical measurements and hence all aspects of the metric system. Initially the work of the BIPM concerned the kilogram and the meter.

The first (and only) follow-up comparison of the national length standards with the international prototype was carried out between 1921 and 1936, and indicated that the definition of the meter was preserved to within 0.2 μm . At this time, it was also decided that a more formal definition of the

⁴ This could have been possible with a second pendulum, where an accurate clock suffices to reconstruct the 'meter'.

⁵ One could argue that nothing was won, given that the meter standard still depended on an artifact. The point however is that a new standard, with exactly the same value as the old one, could be established at any time by repeating the experiment of Delambre and Méchain.

⁶ Argentina, Austria-Hungary, Belgium, Brazil, Denmark, France, Germany, Italy, Peru, Portugal, Russia, Spain, Sweden and Norway, Switzerland, Ottoman Empire, United States of America, and Venezuela. (Note the absence of the UK, in those days still a dominant player on the international stage. However, the UK joined the BIPM soon after, namely in 1884.)

⁷ BIPM = Bureau International des Poids et Mesures.

⁸ CGPM = Conférence Générale des Poids et Mesures.

⁹ Because the prototype is a line standard, its full length is 102 cm, slightly longer than 1 meter. Cross-sectionally, it measures 16 mm $\times$ 16 mm.

meter was required. The 1889 decision had stated merely that the "prototype represented the metric unit of length at the temperature of melting ice. The new definition was formally agreed at the 7th CGPM in 1927. It reads as follows: *"The unit of length is the meter, defined by the distance, at 0°, between the axes of the two central lines marked on the bar of platinum–iridium kept at the Bureau International des Poids et Mesures and declared Prototype of the metre by the 1st Conférence Générale des Poids et Mesures, this bar being subject to standard atmospheric pressure and supported on two cylinders of at least one centimeter diameter, symmetrically placed in the same horizontal plane at a distance of 571 mm from each other."*¹⁰

The prototype meter, being an artifact, remained subject to potential loss or damage. Theoretically the experiment of Delambre and Méchain could be replicated to produce a new prototype. However, this would come at a considerable cost. Moreover it is quite unlikely that the precise circumstances of the original experiment could be repeated¹¹. Hence the new standard would certainly differ from the old one.

By the 2nd half of the 20th century, interferometry advances offered a more convenient way to reproduce the standard of length measurements, both more practical and precise. In particular de noble gas krypton (and more precisely its ⁸⁶Kr isotope) proved to be a useful source of light for this purpose. Krypton-86, like all noble gases, has a large number of spectral emission lines, but it is famous for one specific, highly stable orange-red line with a corresponding wavelength of 605,78 nm in a vacuum.

Accordingly, the 11th CGPM in 1960 agreed a new definition of the meter¹²: *"The metre is the length equal to 1 650 763,73 wavelengths in vacuum of the radiation corresponding to the transition between the levels 2p₁₀ and 5d₅ of the krypton 86 atom."* In this way the precision further improved to 0,01 µm.

The same year 1960 also sees the first working of the laser, an optical device characterized by the emission of a narrow, highly concentrated, and intense beam of light. Laser light is therefore highly monochromatic, and is also coherent (all the light has the same phase), both of which are advantageous for interferometry. Laser technology allowed for a very precise determination of the speed of light, namely $c = 299\,792\,458$ m/s, value approved by the 15th CGPM in 1975. And soon after, this achievement led to yet another definition of the meter. The 17th CGPM of 1983 stated: *"The metre is the length of the path travelled by light in vacuum during a time interval of 1/299 792 458 of a second."* With this definition, the precision further improved to 0,1 nm (1 Å).

The fundamental novelty of this definition is that the unit of length depends on another physical quantity, the second, unit of time. To emphasize this point, the 1983 definition was reworded in 2019 in the 9th edition of the SI Brochure (published by the BIPM): "The meter, symbol m, is the SI unit of length. It is defined by taking the fixed numerical value of the speed of light in vacuum c to be 299 792 458 when expressed in the unit $\text{m}\cdot\text{s}^{-1}$, where the second is defined in terms of the caesium frequency $\Delta\nu_{\text{Cs}}$ ". This definition is known as the caesium standard. The adoption of this standard is

¹⁰ These support locations are at the so-called Bessel points of the prototype. The Bessel points of a bar are the specific positions for two supports, placed symmetrically, that minimize the length change of the bar due to gravity-induced deflection. Named after Friedrich Wilhelm Bessel, these points are primarily used in metrology to maintain the highest accuracy when measuring line standards.

¹¹ Let alone the fact that the dimensions and shape of the Earth are not absolutely constant either.

¹² The principle behind this definition, using the wavelength corresponding to a spectral line, was not new, as the same approach had been used many years earlier (1927) for the definition of the ångström as a secondary unit of length for spectroscopic measurements ($1\text{ Å} = 10^{-10}\text{ m}$). In that case the red line of Cadmium was used.

motivated by the fact that time can be measured more accurately than length and that any (laser) light source of known frequency can be used to realize a meter.

In the caesium standard the second is being defined as “the duration of 9 192 631 770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the caesium 133 atom”. In formulas, the fundamental properties of the ^{133}Cs hyperfine transition radiation, are given by:

- Frequency: $\Delta\nu_{\text{Cs}} = 9\,192\,631\,770\text{ Hz (s}^{-1}\text{)}$.
- Period: $\Delta t_{\text{Cs}} = \frac{1}{\Delta\nu_{\text{Cs}}} = \frac{s}{9\,192\,631\,770}$.
- Wavelength : $\Delta\lambda_{\text{Cs}} = c \cdot \Delta t_{\text{Cs}} = \frac{299\,792\,458}{9\,192\,631\,770} m$

Resulting in the relations between the second and the meter and the caesium radiation characteristics.

- $1\text{ s} = 9\,192\,631\,770\ \Delta t_{\text{Cs}}$
- $1\text{ m} = \frac{9\,192\,631\,770}{299\,792\,458} \Delta\lambda_{\text{Cs}}$

With this definition, the story of the meter reaches a provisional endpoint. We have an unambiguous definition that allows the meter value to be reproduced with extreme accuracy in a practical manner. In practice, we fall back on derived standards in the form of gauge blocks¹³.

A gauge block (aka Jo block) is a metal (e.g., hardened steel alloy or tungsten carbide) or ceramic block with two opposing faces ground and lapped precisely flat and parallel, a precise distance apart. Gauge blocks are the main means of length standardization used in industry. They are used as a reference for the calibration of measuring equipment in machine shops. Their origin goes back to 1896 (Swedish machinist Edvard Johansson). An important feature is that they can be joined together with very little dimensional uncertainty. They are joined by a sliding process (called wringing) which causes their ultra-flat surfaces to cling together.

Gauge blocks are available in various grades, depending on their intended use. For instance, the ISO 3650 standard for the design and metrological specifications of rectangular gauge blocks from 0,5 mm to 1000 mm in length:

- Calibration (K) : for calibration of inspection gauge blocks or high precision gauging equipment
- Inspection (0) : used by inspection departments to calibrate precision tools and check measuring instruments
- Workshop (1) : used as shop standards for setting up machine tools and checking fixtures
- General purpose (2) : used for routine daily measurements and general shop floor gauging

Often gauge blocks are produced in large batches. After grinding and fine lapping, all blocks are precisely measured. Only those that accidentally approach the theoretical size, are certified as K.

Measuring gauge blocks is done with a laser interferometer (usually an iodine-stabilized laser). The block is wrung onto a flat reflective plate. Monochromatic laser light (with a known wavelength λ) is shone on it, creating an interference pattern of dark and light fringes. The distance L between the

¹³ Dutch: eindmaten.

parallel faces of the block is calculated by counting the number of fringes N , using the formula: $L = \frac{N \times \lambda}{2}$.

This measurement is performed under strict conditions: in a climate-controlled room at a temperature of exactly 20,00 °C with sensors that compensate the result for air pressure and humidity.

Blocks are kept lightly oiled and are stored and used in dry climate-controlled conditions. Uncoated steel gauge blocks can last for decades without rusting.